

Lab 04 Reflective Journal – Exploratory Data Analysis

In this lab, I learned more about Exploratory Data Analysis (EDA) and why it is important before working with machine learning. Before this lab, I mostly thought analyzing data meant looking at the values in a dataset. After working with the Titanic dataset, I learned that we can use statistics and visualizations to understand the data better and find patterns that may not be obvious at first.

One thing I learned was how to first check and understand a dataset. I used commands like `df.head()`, `df.info()`, and `df.describe()` to look at the data. I was able to see the different columns, data types, missing values, averages, minimums, and maximums. For example, the dataset showed an average passenger age of about 29.7 years and an overall survival rate of 38.4%. This helped me understand that checking the data should come before trying to make conclusions from it.

The visualizations were the most helpful part for me. By using countplots and histograms, I could clearly see differences in survival based on passenger class, gender, and age. The graphs showed that first-class passengers had a better chance of surviving than third-class passengers. They also showed that females survived at a much higher rate than males and that young children had a noticeable survival pattern. Seeing these patterns on a graph was much easier for me than trying to understand them only from numbers.

After completing two experiments on my own, I also noticed that the first compared the port where passengers boarded with survival. I found that passengers who boarded at Cherbourg had a higher survival rate, while more passengers died than survived at Southampton and Queenstown. For the second experiment, I used a boxplot to compare ticket fare and survival. The graph showed that passengers who survived generally paid higher fares. I also learned that this does not mean paying a higher fare directly caused someone to survive. There could be other factors, such as passenger class, connected to the result.

Overall, this lab helped me understand why EDA is an important part of working with data. I learned how different plots can be used for different types of information and how visualizations can show patterns, differences, and outliers that may be difficult to notice from summary statistics.

alone. I feel more comfortable reading graphs now and understanding what the data is actually showing instead of only running the code.